

Measuring the Urban-to-Wilderness Domain Gap in Aerial Person Detection on Raspberry Pi 4 Hardware

Abstract—The only aerial corpora large enough to train a person detector are urban traffic surveillance, while the missions that matter are flown over wilderness. Almost every published accuracy figure in this field is measured in-domain, so the cost of that mismatch is unknown. We train tiled nano-scale detectors on VisDrone and evaluate them zero-shot on WiSARD, a wilderness corpus of 37,058 tiles from 38 flight sequences, with intervals resampled over sequences. Person detection falls from 0.3997 to 0.2567 AP@0.5. A resize-only control scored with inference-time slicing reaches 0.0449, bounding the gain from training-time tiling at $5.7\times$. Contrary to the usual small-object framing, wilderness targets are three times larger than the training distribution and are viewed from nadir. Halving the inference input recovers 0.068 AP@0.5, but by suppressing 46% of false positives rather than by finding more people. Two training-time factors are hidden by in-domain evaluation: cross-domain accuracy is not monotone in training, peaking near epoch 70 and falling by roughly a sixth while in-domain validation still rises, so the gap we report is an upper bound. Raising the negative-tile fraction from 1.33% to 9.55% cuts false positives by a third at mid-range matched recall, though not at every level. Fine-tuning recovers a factor of 6.1, saturating near sixteen labelled sequences, and repeated coverage lifts encounter-level detection from 0.609 to 0.923. Throughput is measured on a Raspberry Pi 4 without non-maximum suppression, so it bounds deployed cost from below; no result here is flight-validated.

Index Terms—aerial person detection, search and rescue, domain shift, small-object detection, edge inference, unmanned aerial vehicles

I. INTRODUCTION

Search and rescue is a race against time, and the ground to cover is often too large or too dangerous for a walking team. A small quadcopter with a camera sweeps the same ground in a fraction of the time, reaches terrain a team on foot cannot, and returns imagery a coordinator can act on, if its onboard computer is cheap enough to afford, and a Raspberry Pi 4 costs a fraction of a Jetson-class board while drawing a few watts. Prior work has flown this hardware class, in Pixhawk-based Pi 4 payloads, low-cost detection and geolocation pipelines, and an ArduPilot avalanche search drone [1]–[3], cited here for the hardware class, not as baselines. We assembled a payload of the same class around a Pixhawk 2.4.8 and a Raspberry Pi 4 (Fig. 2) as the deployment target, though every measurement below is made on recorded imagery and on that Pi 4 on a desk.

The main difficulty for such a platform is not model capacity, it is data: the only aerial datasets large enough to train a person detector are urban traffic-surveillance corpora,

while the missions that matter are flown over wilderness. A detector must be trained in one visual domain and flown in another, and the literature says little about what that costs, because almost every published search-and-rescue detection number is measured in-domain. A thermal camera beats a visible one here, at 0.977 against 0.892 mAP@0.5 on the same scenes; we work in RGB because payload, cost and power leave this airframe class little choice, and the visible-spectrum gap decides whether thermal is worth buying.

Three kinds of prior work touch this problem, and each stops short of it. Tiled inference such as SAHI [4] rescues small targets but is reported as an inference-time wrapper, measured in-domain. Cross-domain drops have been measured for aerial detectors [5]–[7], but as single before-and-after numbers, and edge quantisation studies report the speed and accuracy trade on Pi-class hardware [8], [9], but for detectors that never leave their training domain. No study connects the three.

This paper makes that measurement, training tiled nano-scale detectors on VisDrone, a three-class urban aerial corpus, and evaluating them zero-shot on WiSARD, a wilderness corpus of 37,058 tiles from 38 flight sequences. Person AP@0.5 falls from 0.3997 in-domain to 0.2567 across domains, and the shift is not the one usually assumed: WiSARD persons are larger and seen from nadir. The main contributions are:

- A sequence-resampled measurement of the urban-to-wilderness gap for a Pi-class detector, and a characterisation of the shift as one toward larger, nadir targets, without ranking the size and viewpoint components (Sections III and IV).
- Evidence that tiled training preserves most of what survives the domain change, worth at most $5.7\times$ over a resize-only control, and that halving the inference input recovers 0.068 AP@0.5 by suppressing 46% of false positives rather than finding people (Section IV).
- Two training-time effects the in-domain view hides: cross-domain accuracy peaks near epoch 70 then falls by a sixth while in-domain validation rises, and extra negative tiles cut false positives at mid-range matched recall by a third (Section V).
- A bound on what target data buys: fine-tuning recovers a factor of 6.1, saturating near sixteen labelled sequences (Section IV).

Section II reviews prior work, Section III the corpora, plat-

form, pipeline and metrics, Section IV the results, Section V their interpretation, Section VI their bounds, and Section VII concludes.

II. RELATED WORK

A. Aerial Person Detection for Search and Rescue

Nano-scale YOLO variants are the practical standard for on-board detection [10], [11], from drowning-swimmer detection [12] to a thirty-three-model sweep across RGB/LWIR fusion ratios [13]. A recent survey finds most aerial personnel objects under twenty pixels and in-domain VisDrone pedestrian accuracy below 50% [14]. We train on VisDrone [15] and evaluate on WiSARD [16].

Two properties of this literature affect how its numbers should be read. First, published results are overwhelmingly in-domain: the WiSARD visual baseline of 0.892 mAP@0.5 was measured after removing every tile with no person from both training and test, removing most chances for a false positive. Second, the protocol is worth more than many method contributions [17]. The same detector scores AP 80.1 under VOC2012 and 91.5 under SAR-APD, and one recent system counts a detection correct at IoU 0.20 and selects on F1 [18]. Our numbers use a strict COCO-style IoU 0.5 over every tile, including the 85.8% with no person. That has a measurable cost. Rescoring one fixed detection set under the corpus authors' positives-only protocol raises AP@0.5 from 0.257 to 0.329, because 77.1% of our detections fall on the 31,793 empty tiles that protocol discards. Protocol thus accounts for 11.3% of the apparent distance to 0.892; the remaining 0.563 is domain shift.

B. Tiling and Small-Object Inference

Resizing a high-resolution aerial frame to a detector's input destroys small targets before the network sees them. Slicing the frame instead is the established remedy. SAHI [4] formalises it and reports AP gains of 5–7% from sliced inference alone, more with slicing applied during fine-tuning. A stride-4 detection level attacks the same problem architecturally, lifting mAP@0.5 from 0.786 to 0.849 at unchanged parameter count [5]. We adopt tiling at training time rather than as an inference-only wrapper, and treat the untilted resize as a control.

C. The Cross-Domain Gap

Where cross-domain transfer has been measured in aerial detection, the drops are large and consistent. One study reports mAP@0.5 falling from 0.786 in-domain to 0.377–0.398 on an unseen corpus [5], and another observes 0.94 to 0.32 for an aerial infrared detector moved between two infrared datasets [6]. The most directly comparable pair comes from [7]. YOLOv8n scores mAP@0.5 of 35.9 transferred from COCO to the SARD wilderness corpus and 86.8 after fine-tuning, a gap the authors attribute to SARD's bird's-eye view. A deployed aerial-ground search system fine-tunes YOLOv11 on VisDrone with class remapping [19], a precedent for our training corpus, and domain adaptation is named as one of four open challenges for the field [14]. Those studies report the gap

as a single before-and-after number; this paper decomposes it into what tiling preserves, what inference resolution recovers, what schedule and negative sampling contribute, and what target-domain data closes, for a detector that fits a Pi 4.

D. Quantisation on Raspberry Pi Class Hardware

Integer quantisation trades accuracy for speed on CPU-class hardware [20]–[22]. A Raspberry Pi ARM CPU sees a $2.61\times$ INT8 speedup with no accuracy loss except under static quantisation [8], the scheme an edge deployment would use. The detector case is harsher. YOLOv8n loses 13.5 points of mAP@0.5 from FP16 to INT8 against 5.8 for YOLOv8s [9], and INT8 YOLOv4-tiny runs on a Pi 5 at 28.2ms and 13.85 W [23]. We reproduce the nano-scale result on a Pi 4 and locate the loss in the integer-activation path.

III. METHOD

A. Problem Setting

A rescue team flies a consumer-class quadcopter with a visible-spectrum camera at 20–120 m over wilderness. The onboard computer is a Raspberry Pi 4 with no accelerator. The detector must be trained on public data, because no wilderness corpus is large enough to train from. Everything below is measured on recorded imagery and on a Pi 4 on a desk (Fig. 1).

B. Target Platform

The detector is intended for a quadcopter we assembled for this purpose. Fig. 2 shows the airframe and its labelled components: a Pixhawk 2.4.8 flight controller, a Raspberry Pi 4 companion computer, a Camera Module V2, an M8N GPS, an X6B receiver, 40 A ESCs, 920 Kv brushless motors, 1045 propellers and a 3S LiPo battery. The Pi 4 is the only compute on board and carries no accelerator, which is what fixes the latency budget of Section III-H. The platform defines the deployment target and is not itself part of the measurement: every accuracy figure below comes from recorded imagery and every latency figure from a Pi 4 on a desk, so nothing reported here is flight-validated (Section VI).

C. Corpora and Label Space

Training used VisDrone2019-DET [15], an urban aerial corpus of 8,629 annotated frames in the official 6,471/548/1,610 train/validation/test-dev partition, from 208, 76 and 61 video sequences. Its ten categories were collapsed to the three a search-and-rescue operator acts on (person, vehicle, two-wheeler), giving 343,205, 38,759 and 75,102 instances per split; categories outside the mapping were discarded, not merged into a background class.

Cross-domain evaluation used WiSARD [16], a wilderness corpus recorded from consumer airframes at 20–120 m. We used its visible-spectrum subset at every tenth frame, giving 2,586 frames across 38 sequences. WiSARD is annotated for person only, so we declare it in the same three-class space and report the person row exclusively; any correct vehicle detection would otherwise count as a false positive against absent ground truth.

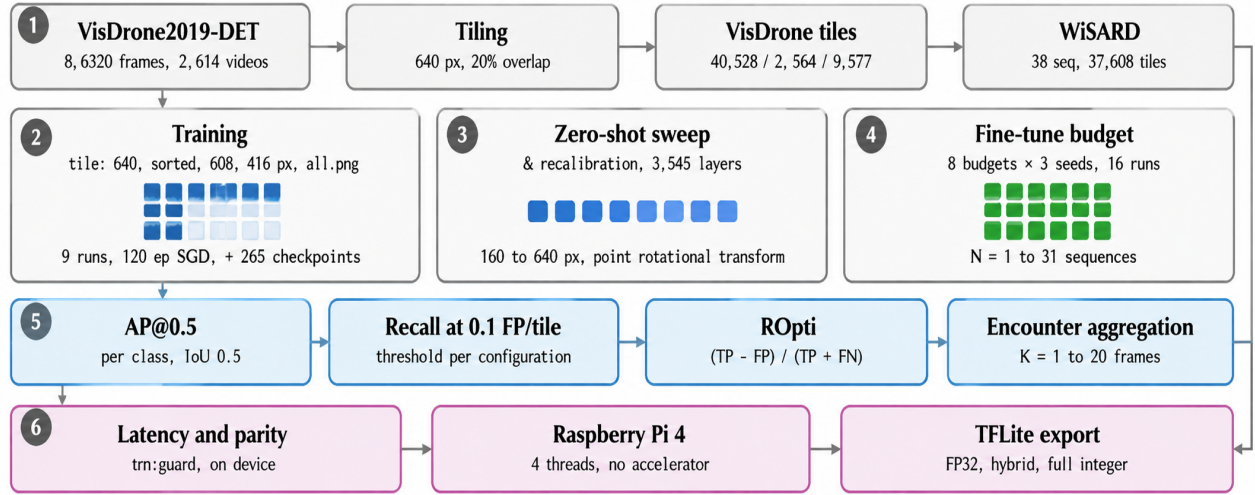


Fig. 1. The measurement pipeline, numbered by its five stages. Each chip is one run, and the pale cells in the training grid mark the three configurations that ran at one seed rather than three. The on-device branch is fed from the trained checkpoint, not from the metrics above it. Every count is read from the committed result files.

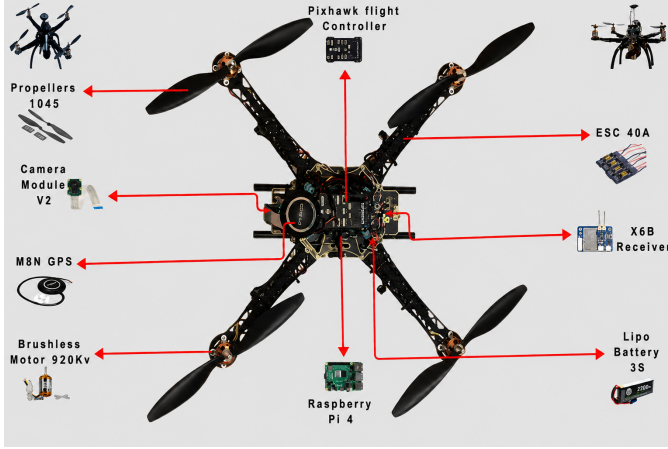


Fig. 2. The assembled quadcopter and its labelled components: Pixhawk flight controller, Raspberry Pi 4, Camera Module V2, M8N GPS, X6B receiver, 40 A ESC, 1045 propellers, 920 Kv brushless motor and 3S LiPo battery.

Frames within one flight are near-duplicates, so a frame-level partition inflates accuracy, and all partitions are made at sequence granularity. The converted VisDrone layout restarts video identifiers per split, so we parse the identifier from the filename and namespace it by split; MD5 hashing over all 8,629 images found zero cross-split duplicates.

D. Tiling

Tiling is applied when the dataset is built, not only at inference, because resizing destroys small targets before training as well as before detection. Each frame was cut into overlapping 640px tiles at native resolution with 20% overlap, the final tile in each row and column flushed to the far edge. A ground-truth box was kept for a tile when at least 30% of its area fell inside. All positive tiles were kept and negatives subsampled at 17.5% per frame, giving 40,523, 2,784 and 9,377 tiles. Applied per frame, that fraction rounds away most frames' few empty tiles, so the realised composition is 1.33% negative, not the

15–20% intended. As a control, an otherwise identical model was trained on whole frames resized to the same input.

The class balance differs sharply between the corpora, and it matters below. Of the 40,523 VisDrone training tiles, 39,983 (98.7%) contain at least one annotated object; of the 37,058 WiSARD tiles, only 5,265 (14.2%) contain a person.

E. Training

We trained YOLO11n and YOLOv8n for 120 epochs at 640px with batch 32, and one YOLO11n variant at 416px with batch 48, all under SGD with a cosine schedule, early-stopping patience of 25, deterministic kernels at seed 42, and COCO-pretrained initialisation, on a single 8 GB RTX 3060 Ti. The two headline configurations were replicated at seeds 123 and 456.

F. Target-Domain Fine-Tuning

To bound what target-domain data recovers, the WiSARD tiles were partitioned into sequence-disjoint train, validation and test sets. Allocation balanced annotated person boxes rather than tile counts, since six of the 38 sequences contain no person and per-sequence box counts range from 0 to 1,122, so a tile-balanced split can put almost every person on one side while looking even. The realised split holds 31, 2 and 5 sequences with 5,685, 1,161 and 1,499 boxes, or 68.1/13.9/18.0% of the annotated people. Fine-tuning initialised from the VisDrone-trained tiled 640px checkpoint under SGD at a reduced learning rate of 10^{-3} . Validation accuracy peaked at epoch 5 and did not improve over the next sixteen, so we report the epoch-5 checkpoint.

G. Metrics and Uncertainty

Detections were matched greedily in descending confidence at IoU 0.5, each ground-truth box consumable once so duplicates count as false positives, and average precision is all-point interpolated. Because a missed person costs far more

than a dismissed false alarm, we also report recall at a fixed budget of 0.1 false positives per tile, with the threshold that achieves it. Generic metrics understate the operational cost of false alarms here [17], so we additionally report $\text{ROpti} = (\text{TP} - \text{FP})/(\text{TP} + \text{FN})$ [7], unbounded below and threshold-dependent; we give it at the operating point above and at its own maximum, naming the threshold each time.

Confidence intervals resample sequences, not tiles or boxes. WiSARD tiles come from consecutive video and are strongly correlated within a flight, so a tile-level bootstrap would treat near-duplicates as independent and return an indefensibly narrow interval. All intervals are 10,000-resample percentile intervals over the 38 sequences; where two configurations are compared, both are scored on the same resampled sequences and the interval is taken on their difference, the right test for a paired comparison.

H. On-Device Measurement

Deployment measurements were taken on the Raspberry Pi 4 under TensorFlow Lite with four threads, discarding 60 warm-up iterations before 160 timed ones (100 before 200 for the hybrid-integer export), with thermal throttling checked before and after each run. Two caveats attach to every latency reported. The exported graph emits raw predictions, so the timed loop covers decode, letterbox, forward pass and dequantisation but not non-maximum suppression, making each figure a lower bound. The governor was performance for the 640px exports against ondemand for the 416px ones, so the two sets are not directly comparable.

Availability. Both corpora are public. Training used Ultralytics 8.3.253, PyTorch 2.5.1, CUDA 12.4 and Python 3.12.3; the Pi used `tflite-runtime` 2.13.0 under Python 3.9.2. Code, pinned dependencies and the per-detection caches behind every table will be released on publication.

IV. RESULTS

A. Tiling Accounts for Most of What Survives the Domain Change

Table I reports person-class accuracy in both domains. In-domain, tiling raises person AP@0.5 from 0.2390 to 0.3997. Over wilderness imagery the contrast becomes decisive, the tiled model reaching 0.2567 against the resize-only control’s 0.0449, a factor of 5.7. The control was trained on resized whole frames but scored on native tiles merged with cross-slice suppression, our implementation of the inference-time slicing recipe SAHI [4] publishes. The contrast is therefore training-time tiling against inference-time slicing, and $5.7\times$ is an upper bound, since the control is scored outside its own training regime and the raw frames needed to score it inside are no longer held. Tiling is also the only cross-domain row that separates. YOLO11n at 0.257, YOLOv8n at 0.265 and the 416px variant at 0.225 all fall inside one another’s intervals, so only tiling is resolved at this sample size. Replicating both headline rows at seeds 123 and 456 leaves that intact, every tiled seed (0.2567, 0.2536, 0.2868) beating every control seed

TABLE I
PERSON-CLASS DETECTION ACROSS DOMAINS. IN-DOMAIN ROWS ARE SINGLE EVALUATIONS. INTERVALS ARE 95% PERCENTILE INTERVALS OVER 38 RESAMPLED SEQUENCES. THE THREE BLOCKS USE DIFFERENT EVALUATION SETS AND ARE NOT COMPARABLE ACROSS BLOCKS. BEST PER BLOCK IN BOLD. AP@0.5 OF 0.2–0.5 IS TYPICAL FOR SMALL-OBJECT AERIAL PERSON DETECTION ON CORPORA SUCH AS VISDRONE [15], REFLECTING THE DIFFICULTY OF THE TASK AT ALTITUDE RATHER THAN A WEAK DETECTOR.

Configuration	AP@0.5 [95% CI]	Recall [95% CI]
<i>In-domain, VisDrone test-dev</i>		
YOLO11n tiled 640	0.3997	0.356
YOLOv8n tiled 640	0.3978	0.351
YOLO11n tiled 416	0.2898	0.262
Untiled 640, control	0.2390	0.208
<i>Zero-shot, 38 sequences, 37,058 tiles</i>		
YOLO11n tiled 640	0.257 [.141–.393]	0.320 [.217–.416]
YOLOv8n tiled 640	0.265 [.145–.395]	0.323 [.224–.413]
YOLO11n tiled 416	0.225 [.102–.368]	0.288 [.177–.390]
Untiled 640, control	0.045 [.010–.124]	0.093 [.030–.178]
Tiled 640, infer. @320	0.329 [.196–.472]	0.386 [.281–.489]
External, VisDrone-pre. [24]	0.041 [.011–.096]	0.080 [.025–.144]
<i>Held-out target domain, 5 sequences, 1,499 boxes</i>		
Tiled 640, zero-shot	0.0746	0.120
+ fine-tuned	0.4575	0.459

(0.0449, 0.0427, 0.0267). The ratio of means is 7.0 and the seed spread is 13–16% of the interval width.

The absolute figure is still low. At 0.1 false positives per tile the model recovers only 32.0% of annotated people, and at frame granularity, where the budget is fourteen times tighter, recall is 11.3%.

Under ROpti the picture is harsher. At the 0.1 FP/tile operating point ROpti is -0.12 , so the detector emits more false positives than true positives, which plain recall hides. At its own optimum it is 0.086 [.012–.205] against the 0.09 [7] reports for the analogous zero-shot transfer, and the untiled control reaches 0.004 [.002–.041].

B. The Shift Is Toward Larger, Nadir Targets

Fig. 3 shows the two corpora in identical 640px tile space, differing in the opposite direction to the one the small-object literature suggests. VisDrone training persons average 20.88px across 218,769 instances, with 74.8% between 8 and 32px, while WiSARD persons average 67.30px across 8,345 instances, with 79.5% at or above 32px. The bounding-box aspect ratio shifts from 2.00 to 1.46, consistent with a change from oblique street-level views, where a standing person is an upright figure, to the near-nadir view of a wilderness overflight, where the same person is a compact top-down blob. The cross-domain target is therefore roughly three times larger than the training distribution, not smaller.

C. Reducing the Inference Input Helps, but Not for the Expected Reason

The model is fully convolutional, so apparent object size can be changed at inference by altering the input resolution, with no retraining. Sweeping it from 640 down to 160px gives a smooth inverted-U in cross-domain accuracy, peaking at 320px, where AP@0.5 reaches 0.3291 against 0.2567 at

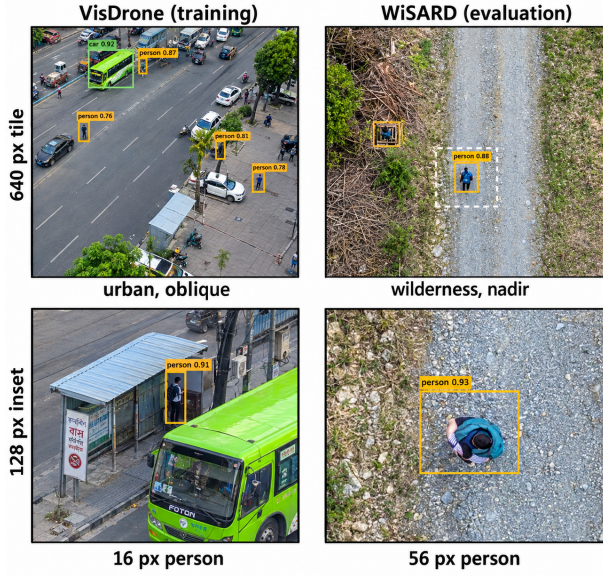


Fig. 3. The domain shift, at identical scale. Top: 640 px tiles cut at native resolution, with annotated persons boxed. Bottom: the same 128 px window around a person of each corpus’s median size, a 16 px oblique figure against a 56 px top-down form.

the native 640. A paired sequence-level bootstrap puts the difference at $+0.068$ with a 95% interval of $[0.019, 0.117]$, significant between 512 and 320 px and indistinguishable from zero at 256 px and below. It survives Bonferroni correction across the seven comparisons (marginally at 320 px, $+0.002$, against $+0.028$ at 512).

The mechanism is not the one the size measurement suggests. Decomposing the change shows true positives rise only from 5,081 to 5,202 and the recall ceiling moves from 0.609 to 0.623, while false positives fall from 204,087 to 109,493, a 46% reduction. Halving the inference resolution does not reveal people the detector was missing, it stops it asserting people who are not there. The gain is real, significant and free, but it is a precision effect.

Requiring agreement between 640 and 320 px removes 47–64% of false positives but also 7–9% of true positives, and no consensus variant beats single-scale 320 (0.3269 against 0.3296), so what is lost at reduced resolution is not only spurious.

D. What Target-Domain Data Recovers

Fine-tuning on 31 wilderness sequences and testing on 5 the model never saw raises person AP@0.5 from 0.0746 to 0.4575, and recall from 0.120 to 0.459, a factor of 6.1. Zero-shot on this subset scores 0.0746, against 0.2567 over all 38 sequences, so these five are harder than the corpus average and the multiplier is specific to them.

The amount of target data needed saturates. We repeated the fine-tune at 1, 2, 4, 8, 16 and 31 training sequences under three seeds. Mean person AP@0.5 is 0.082, 0.189, 0.372, 0.369, 0.456 and 0.449. The $N = 31$ point carries no seed variance by construction: thirty-one sequences exhaust the pool, the three seed splits are byte-identical, and deterministic training

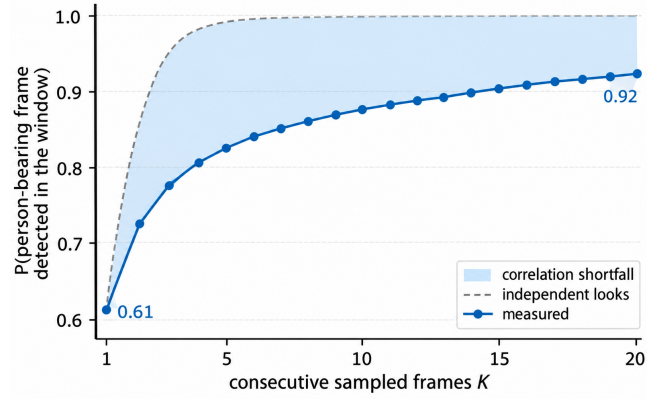


Fig. 4. Encounter-level detection against window length K , with the independent-looks bound above it. The shaded band is the shortfall between the two.

returns one number three times. Budget runs trained for 15 epochs while the headline fine-tune early-stopped at epoch 21 of 40, which is why the curve reads 0.449 at 31 sequences against the headline’s 0.4575. Below that, spread matters more than mean: the seeds range over 0.347 at $N = 4$, because per-sequence annotation counts run from 0 to 1,122 and a small draw is decided by which sequences it contains. At sixteen the spread collapses to 0.035 and the mean matches the full set, so at this sample size further labelling brings no measurable gain, and a modest annotation campaign buys most of the available recovery.

E. Repeated Coverage Recovers Part of the Shortfall

Per-frame recall is not what a search cares about. An aircraft passes over the same ground repeatedly and a person need only be found once. WiSARD carries no track identifiers, so we measure frame presence within a flight. A frame is positive when any tile holds an annotated person and detected when any tile yields a true positive above the operating threshold. A window of K consecutive positive frames counts as detected if any frame in it is. The windows come from ground truth, so this bounds what repetition can recover rather than describing a live search.

This study is scored at 320 px with an operating threshold of 0.1515, the reduced-resolution row of Table I, not the 640 px headline. Over 1,703 positive frames in 32 sequences, a single frame is detected with probability 0.609 [0.472–0.742], rising to 0.923 [0.799–0.997] over 20 consecutive sampled frames (Fig. 4). Tiles were built at every tenth frame, so K counts sampled frames about a third of a second apart at 30 fps, making $K = 20$ roughly 6.7 s of flight. The curve sits below the independent-looks bound, by up to 0.178 at $K = 4$, because consecutive views share pose, illumination and background.

F. Export Precision on the Target Hardware

Latency was measured on the Pi 4 at four threads as described in Section III-H, so every figure here excludes non-maximum suppression and bounds deployed cost from below. TFLite FP32 runs at 495 ms and the hybrid-integer export at

761 ms, both within 0.72 percentage points of the PyTorch reference on all-class mAP@0.5, not comparable to the person-only numbers above. That parity rules out a preprocessing mismatch. Identical letterbox, channel order and input scaling give faithful results everywhere except the integer-activation path. Full-integer quantisation, where activations as well as weights become integers [20], [22], runs at 410 ms, a $1.21\times$ speed-up, and costs 12.80 percentage points. The split between the two integer exports locates the fragility in the activations rather than the weights, what a Softmax inside the distribution-focal-loss head would produce, and an independent report reaches the same attribution [5]. The hybrid variant is slower and no more accurate, so it is dominated outright.

V. DISCUSSION

The false-positive result has an explanation in the class balance reported above. The detector was trained almost entirely on evidence that something is present, then asked to spend deployment deciding that nothing is. Retraining identically on every negative tile VisDrone can supply lifts the negative fraction from 1.33% to 9.55%, but the benefit depends on where recall is fixed. Cross-domain false positives at matched recall change by +12.0, -6.6, -32.9, -41.9 and -32.6 percent at recall 0.10 through 0.50, so the lowest recall point moves the wrong way. AP@0.5 rises from 0.2570 to 0.2784 with the recall ceiling unchanged within 0.006. Three bounds apply. This is one baseline checkpoint against one control, so there is no seed axis. The corpus exhausts at 9.55% against WiSARD’s 85.8%. The same comparison at epoch 50 shows no benefit, placing the effect in late-training divergence.

The same recall-first framing explains why full-integer quantisation is a poor trade here. Our 12.8-point cost for a $1.21\times$ speed-up matches the nano-scale figure in [9], against 5.8 points for a small-scale model, and on the device the trade is unnecessary too. Moving the deployed payload to floating point raises median inference from 645 to 772 ms, a factor of 1.20, yet across sessions of 6 to 17 minutes both precisions sustain the same $0.91\text{--}1.20\text{ s}^{-1}$ end-to-end. Capture, tiling and encoding set the pace, so a 20% slower model is absorbed rather than paid for, and recovering the 12.8 points cost no throughput.

Throughput depends on what is timed, and on what governor is set. Over stored frames a single 416px tile costs 171 ms under `ondemand`, against `performance` behind the 640px figures, so the two are not a like-for-like pair. The four tiles covering a frame spend 0.69 s in the interpreter and the benchmark completes 1.33 sets per second, while the full service, which also captures and encodes, runs at the rates above. Whether either suffices depends on airspeed and altitude, which we have not measured, and the 320px gain is unconfirmed on-device. The operating point must also be set deliberately. Across all 37,058 tiles the conventional 0.25 threshold recovers 30.1% of annotated people against 38.7% at 0.10, so the inherited default discards 28% of available recall for 0.18 more false positives per tile and no latency cost.

One factor cuts across every number above. Scoring the periodic checkpoints of both baseline seeds against WiSARD shows cross-domain accuracy is not monotone in training: it peaks at epochs 70 and 60, at AP@0.5 0.3149 and 0.2966, then falls to 0.2570 and 0.2532 by epoch 120, a relative loss of 18.4% and 14.6%, while in-domain validation rises throughout. The checkpoint ordinary practice selects is not the one that transfers, so our reported figures, taken at epoch 120, are pessimistic by roughly a sixth and bound the gap a target-aware stopping rule would leave. This also explains why the negative-fraction control helps only late: the extra negatives mostly prevent this decline rather than raising the peak.

VI. LIMITATIONS

Scale and viewpoint shift together and our experiments cannot separate them. Reducing the inference input changes apparent object size while holding the ground footprint fixed, not ground-sample-distance matching, which would need source frames we no longer hold. The residual gap mixes viewpoint, pose, terrain and illumination.

Statistical support rests on resampling 38 flight sequences, correct for correlated video but small, so the intervals in Table I are wide. New configurations were trained at one seed, except the headline rows, the budget curve at three and the epoch sweep at two. The fine-tuning multiplier is a point estimate, and neither it nor the negative-fraction result is re-derived from the epoch-70 checkpoint.

Most importantly, nothing here is flight-validated: every accuracy figure comes from recorded imagery, and every on-device figure from a Pi 4 on a desk. We make no claim about vibration, rolling shutter, motion blur at airspeed, throttling under a fairing, or latency in a real search.

The benchmark has known weaknesses: WiSARD shows targets in bright clothing in open terrain [17], while real missing persons typically seek shelter, so absolute accuracy here should be read as optimistic. We evaluate visible-spectrum only; a thermal channel is measurably stronger here, and thermal tracking systems for search and rescue already exist [25], [26], the obvious upgrade for a capable airframe.

VII. CONCLUSION

A nano-scale detector trained on urban aerial imagery loses roughly a third of its person-detection accuracy over wilderness. Tiled training bounds that loss, worth at most $5.7\times$ over a resize-only control scored with inference-time slicing, and the shift is toward larger, near-nadir targets. Halving the inference input recovers 0.068 AP@0.5 by suppressing 46% of false positives, not by finding more people, and extra negative tiles cut a further third at mid-range matched recall. Repeated coverage lifts encounter-level detection from 0.609 to 0.923. The rest is an appearance gap only target-domain data closes, with fine-tuning recovering a factor of 6.1 and saturating near sixteen labelled sequences.

For teams deploying this class of system: tiling is not optional, inference resolution, decision threshold, negative sampling and stopping epoch are free and underused knobs,

and sixteen labelled target-domain sequences buy more than any of them.

Three extensions follow: a flight trial of the Pi 4 payload to measure motion blur, throttling and latency at airspeed; a target-aware stopping rule, re-deriving the fine-tuning and negative-fraction results from that checkpoint; and a ground-sample-distance matched evaluation separating size and view-point. The airframe has room to develop alongside them: RF telemetry would extend its range and autonomy, together with work on endurance and precise navigation, and running the detector in the loop with the flight controller would let the aircraft react to what it spots while it is still overhead.

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